

Project Logistics – Version 1

Due Dates

Midpoint check: Oct. 15, 11:59 pm

Final Project: Dec. 1, 11:59 pm

Late Policy: Midpoint submission follows the same late policy for homework assignments. For the final submission, late submissions will NOT be accepted.

The project can be large. Please start early.

There will be a midpoint check. You are supposed to implement about half of the lines for your project (500 lines or more), which can pass compilation. This is worth 4% of your course grade.

TA for Project

Please direct your project questions to the TA Muiyang. The email addresses and office hour information are listed on the course front page of Canvas.

Group Size

- 4 persons per team; 3 is acceptable. Please form your project group ASAP.
- It will be great if you find your own project partners. If you need help to find project partners, please follow the announced procedure to get help from Muiyang.
- After you form your project group, one of the members should sign up your project group on Canvas.

Project Submission

- Submit on Canvas for both the midpoint check and the final project.
- One submission per project group.
- Combine all your files into one archive and/or compressed file. An example in UNIX/Linux/MacOS:

```
tar cvf proj1.tar foo.java bar.java
```

Here, `proj1.tar` is the archive file, and `foo.java` and `bar.java` are your source files.

- The zip or tar file should contain all source files, all files needed to compile your program, and all files needed to run your program.
 - It should also contain a **final project report**.
 - Optional: If you use a version control system such as github, generate a text file of the commit history, call it `commit_history.txt`. For instance, you can do something like:

```
git log --oneline --graph --all > commit_history.txt
```

- Your zip file should **not** contain executable files or object files which can be generated by compiling your program.
- To minimize the storage, it should **not** contain any data files for sharing, or any other non-essential files.

Demo

- After the submission, each group will show a demo of their project. This will take the form of 5-8 minutes recorded video. You can submit the video demo on Canvas if the file size is not too large. Otherwise, you can store the video on onedrive@uf and share a link in your submission.
- For the demo, the data file to be exchanged should be at least 20Mbytes. The piece size will be 16384 bytes. The required free disk size depends on how you handle partial temporary files. You may need more than 150Mbytes. Please make sure that you have enough free disk space for your program to distribute a 20 Mbyte file from a peer to five other peers.
- For the demo, you should try to use multiple machines.
- To get a high score, it is very important to meet the requirements described in the implementation specifics. Writing logs is also important.
- More details will come later.

Final Project Report

The report should be roughly 1-3 pages. A template is provided. It should contain:

- names of the group members.
- degree of completion: This can be one sentence if you have completed everything and the code works. Otherwise, please provide more details on what has been done, what's missing, what works and what doesn't work. Your partial grade depends on it.
- the information needed to test your file, such as a short instruction.
- **division of responsibilities**, i.e., who did which part.
- **Answering the question:** Have the group members contributed approximately equally? If not, provide details if you can, such as what you have done and how many hours you have spent on the project.
- AI use disclosure, i.e., what you use AI for.
- Anything else that you feel worth mentioning.

AI Use

You are allowed to interact with any of the Chat tools and learn how to do network programming. If you choose to do so, please do it in an interactive way as a learning process. Each interaction should produce no more than a small piece of code for a specific purpose. For instance, you can ask:

- How to write a simple server or client?
- How to manipulate raw bits in a packet header in Python?

It is fine if you learn how to code different small pieces and you integrate them together into the project code. You shouldn't use a coding agent to write the whole or a big part of the project code.

To show proper AI use, please add extensive comments/documentation in your code using your own language. The goal is to show that you are involved in the code development and understand the code.

Questions

- If you have difficulty understanding the protocol description, you may search for the BitTorrent protocol description on the Internet and study it. There are many sources of information that explain the protocol. Then, you can come back and read the project description again.

Grading

Each project group will receive a common grade first. After that, there may be an upward or downward adjustment for each group member depending on the contribution and the amount of effort spent.

Partial Credits for Project

If your group can't get the project to work, please document what you have done and who has done what in the project report and explain your situation during the demo. This helps each of you to receive partial credits based on your code.

Sample Files

On Canvas, under the direction 'Files/Project', you can find some files related to the project.

Sample Client.java, Sample Server.java: These are self-explanatory.

project_config_file_small.zip, project_config_file_large.zip: Two sets of sample configuration files and data file to be exchanged. One set has a small data file, around 2 MB; the other has a large data file, more than 20MB. You can play with both scenarios. To demonstrate choking, unchoking, and optimistic unchoking, the data file should be at least 20MB, or perhaps larger.

References

There are many online resources for Python, JAVA and network programming. Feel free to use those resources. You might have to browse many and select the ones that are suitable for you.

For Java, the following book is very good.

Elliott Rusty Harold. [Java Network Programming](#), 4th edition. 2013. O'Reilly Media, Inc.

Some chapters of the 3rd edition can be found online. See [here](#).

Academic honesty

You should not have someone else to write your code or copy the code from someone else. But, you are allowed to use the Internet resources and learn how to do network programming in general. Anyone who violates the above rule will get penalized. **Please note that there are tools for us to check the code similarity.**

Manage Team Dynamics

There can be people problems in a group project. Please be aware of such possibilities and try to resolve any issues early. If you wait for too long, there may not be any time left for remedies. Everyone should think about how to manage difficult team dynamics and how to get everyone to contribute. It can be an essential skill for your career.

FAQ

Q1. Can I assume that the subdirectories for peers have already been set up before I run the peer processes?

Yes. You may create subdirectories with proper names before you run your peer processes. That saves you the burden for managing subdirectories in your program. See the relevant discussion in the project description document.

Q2. How many TCP connections should be used between two peers?

The design may vary. For instance, you may use one TCP connection for each pair of peers. Consequently, a peer maintains only one socket for each neighboring peer. However, you may have two threads associated to the socket, one for getting messages from the neighboring peer and the other for sending messages to the peer. The socket will be shared by those two threads.

Q3. How is the ‘have’ message different from the ‘bitfield’ message?

The ‘bitfield’ message is useful to inform a neighboring peer the availability of many file chunks. You can use it at the start of a process. The ‘have’ message is used to inform a neighbor the availability of a single chunk. It is useful for incremental update between the neighboring peers. For instance, when a peer receives a new chunk, it can use the ‘have’ message to inform other neighbors.

Q4. What should I submit for midpoint check?

You are asked to submit 500 lines or more of compiled code. Provide a readme file containing the names of your project team. That’s all. The objective is to make sure you have started working on it.

Q5. Should each project group submit only one copy of the code?

Yes. Only one of your teammates needs to submit the project (midpoint or final) on Canvas. Please list your group members: (i) in a readme file for the midpoint submission; (ii) in the final project report for the final submission.

Q6. The project looks complicated. How should I start?

I recommend you first read the project description document CAREFULLY. It describes a protocol. The whole project may look complicated, but fundamentally, each peer just sends, receives, reads, replies messages through TCP connections based on the protocol described in that document. After several reading, you will have a broad picture of the project in your head.

Q7. Can I use my own machine to run the project code?

During development, it is easier to work on your local machine. You can run several peer processes on your machine. For the demo, you are asked to run your code on different machines.